

Deep Green AI: Energy Efficiency of Deep Learning across Language–Framework Ecosystems

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ABSTRACT

Context. The rapid growth of Deep Learning (DL) has been accompanied by a steep increase in its environmental footprint, and a natural question follows for practitioners: does the choice of language–framework ecosystem carry an energy price? Answering it requires that every ecosystem run the same experiment and be measured the same way. Both requirements are more demanding than they appear, and neither is verifiable from reported energy figures alone.

Objectives. This paper investigates *how the choice of language–framework ecosystem affects the energy efficiency of DL workloads*, and, inseparably from that, *what an apparatus must guarantee before such a comparison means anything*. We adopt a practitioner-centric perspective on the operational energy–time trade-offs developers face when selecting an ecosystem for training and inference, using computer vision image classification as a representative energy-intensive workload.

Methods. We compare two convolutional architectures (ResNet-18 and VGG-16) across 7 language–framework ecosystems — Python with PyTorch, TensorFlow and JAX; C++ with LibTorch; Java with Deeplearning4j; R with `torch`; and Rust with `tch` — on three datasets (Fashion-MNIST, CIFAR-100, Tiny ImageNet). Three methodological commitments distinguish the design. First, a written experiment specification states what “the same experiment” means across stacks, and 57 automated checks enforce it against the source, so a divergence fails loudly instead of quietly changing a number. Second, energy is read from hardware counters — NVML `nvmlDeviceGetTotalEnergyConsumption` for the accelerator and the RAPL package-energy counters for the CPU — with CodeCarbon running over the same window as a second, independent instrument. Third, each configuration is executed as 5 independent interleaved runs with distinct seeds, making the run rather than the epoch the unit of analysis, and every stack records per-epoch accuracy so that energy can be normalised by the quality actually achieved.

Results. The design yields 210 runs and 12600 doubly instrumented measurement blocks. Between-run repeatability is high: the median coefficient of variation of training energy is 0.44 %. Under a common measurement boundary, training energy differs across ecosystems by 7.8 to 19.1 depending on the block, while final accuracy converges to within 1.3 percentage points on Fashion-MNIST — the differences are energetic, not qualitative. The two instruments agree on energy to 0.5 % over the whole campaign — 0.01 % for blocks longer than 30 s, degrading to 1.3 % below one second, where the block is shorter than the sampling interval — so the difficulty with software estimation is narrower and stranger than inaccuracy: the duration CodeCarbon reports beside each energy figure is not the duration the energy was accumulated over, but the phase plus a constant 3.28 s whenever the phase is shorter than about 11 s and the phase itself above that ($R^2 = 0.9980$). Any power or energy–time analysis built on that field therefore understates power by up to 6.7 for sub-second phases while being exact above ten seconds — a bias that falls precisely on the fastest ecosystems and the inference phase.

Replication also exposed a failure a single-run design cannot see: VGG-16 converges to exactly chance accuracy in 12 of 100 runs and never trains, consuming 10.7 % of the campaign’s energy for nothing, in 4 ecosystems independently and never for ResNet-18. Excluding those runs collapses the cross-ecosystem accuracy spread on VGG-16 on CIFAR-100 from 20.2 to 1.3 percentage points — so almost all apparent disagreement between stacks was runs that failed to train, not stacks computing different things.

Conclusions. Ecosystem selection materially affects the energy profile of DL applications, but the size and even the direction of that effect depend on decisions in the measurement apparatus that are invisible in the resulting data. We report the corrected comparison together with the specification, the shared measurement contract, the conformance checker and the raw dual-instrument records needed to audit it.

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1. Introduction

The rapid advancement of Artificial Intelligence (AI), particularly in the domain of Deep Learning (DL), has delivered groundbreaking progress across domains

such as healthcare, natural language processing, environmental monitoring, and autonomous systems. As models grow in size and complexity, their performance continues to improve, often reaching or surpassing human-level capabilities in specific tasks. However, these achievements have come at a growing environmental cost. The training and deployment of state-of-the-art neural networks demand vast computational resources and energy, contributing to significant carbon footprints [1, 2, 3, 4, 5, 6, 7]. Recent studies show that energy usage and emissions vary widely depending on model architecture, dataset size, hardware infrastructure, and even geographic location of execution [8, 9]. In response, standardized benchmarks, such as MLPerf Tiny, have begun to incorporate energy measurements to assess efficiency under realistic deployment conditions [10]. In a world of limited environmental resources, the ever-growing energy required to power AI has become a pressing concern, making sustainability a central dimension of AI research [11, 12].

Rather than framing this trend solely as a cause of concern, the field is now embracing a proactive paradigm, referred to as *Green AI* [12, 13]. This research line promotes the development of AI systems that are not only accurate and effective but also energy-conscious and efficient. Recent surveys have highlighted advances across lightweight architectures, efficient training and inference strategies, model compression, and hardware-aware optimizations [4, 14, 15, 16]. Evidence increasingly points in the direction of energy improvements not necessarily implying a degradation of accuracy [17].

Beyond algorithmic efficiency, research in software sustainability has shown that the choice of programming language itself can have a substantial impact on energy usage [18, 19, 20, 21, 22, 23, 24, 25]. Nevertheless, while both Green AI and the environmental sustainability of programming languages have been independently investigated, their intersection remains largely unexplored. A first step was made by Georgiou *et al.* [26], who analyzed the energy footprint of Deep Learning frameworks within Python, and by Marini *et al.* [27], who studied the impact of programming languages in AI workloads.

For a practitioner, however, a programming language is inseparable from its *ecosystem*. Choosing a language for DL typically implies adopting a specific software stack, including language bindings, frameworks/libraries, runtimes/execution engines, and low-level driver/toolchain dependencies. As a result, in real-world settings it is often more meaningful to evaluate *language-framework ecosystems* as they are adopted in practice, rather than attempting to isolate a “pure” language effect under synthetic, fully standardized conditions.

Yet, to the best of our knowledge, no prior work has systematically examined this ecosystem-level dimension across multiple language-based DL stacks in the context of GPU-based DL, where frameworks, execution engines, toolchains, and dataset complexity can introduce substantial variability [28].

This paper addresses this gap by presenting a large-scale empirical study on *the impact of language-framework ecosystems on the energy efficiency of DL workloads*. To achieve our goal, we consider two widely adopted convolutional neural networks (ResNet-18 and VGG-16) across 7 language-framework ecosystems, spanning five host languages (Python, C++, Java, R and Rust) and the frameworks each of them offers in practice (PyTorch, TensorFlow and JAX; LibTorch; Deeplearning4j; and the `torch` and `tch` bindings), on three heterogeneous datasets (Fashion-MNIST, CIFAR-100 and Tiny ImageNet).

A cross-ecosystem energy comparison, however, rests on two premises that are rarely stated and never visible in the resulting numbers: that every stack is running the same experiment, and that every stack is measured the same way. We found both to be far harder to satisfy than the literature suggests, and we came to that conclusion the expensive way. An earlier campaign of ours, of exactly this shape, was executed carefully and produced an entirely coherent story; auditing it against its own source code afterwards showed that the stacks had drifted apart in learning rate, input shape, evaluation batching and data-loader parallelism, that the measurement tool had been configured five different ways across them, and that two stacks were capable of running the whole workload on the CPU while reporting nothing but a log line. None of this was visible in the energy figures. All of it moved them.

We therefore treat the apparatus as part of the contribution rather than as a preliminary. Three commitments follow. (i) A written experiment specification states what “the same experiment” means across stacks — model, optimisation, data pipeline, backend, measurement and replication — and 57 automated checks enforce it against the source code, so that a divergence fails a check instead of quietly changing a number. (ii) Energy is read from hardware counters, NVML’s accumulated-energy register for the accelerator and the RAPL package-energy counters for the CPU, with CodeCarbon running over the same window as a second, independent instrument; every block therefore carries two readings that can be held against each other. (iii) Each configuration is executed as 5 independent, interleaved runs with distinct seeds, so that the run rather than the epoch is the unit of analysis, and every ecosystem records per-epoch test accuracy, so that energy can be normalised by the quality actually reached rather than assumed equal.

Our results are correspondingly more modest and more defensible than a bare ranking. Between-run repeatability is high — the median coefficient of variation of training energy across 210 runs is 0.44% — so the intervals we report are genuine. Under a common measurement boundary, training energy differs across ecosystems by 7.8 to 19.1 depending on architecture and dataset, with JAX consistently cheapest, while final test accuracy converges to within 1.3 percentage points on Fashion-MNIST across all 7 stacks. The ecosystem effect is real, it is large, and it is energetic rather than qualitative.

Replication earned its cost in a way we did not anticipate. VGG-16 fails to train at all in 12 of 100 runs — converging to exactly chance accuracy and staying there for the full epoch budget — in 4 of the 7 ecosystems independently, and never once for ResNet-18. Those runs consume 10.7% of the campaign’s energy and produce nothing, and they are invisible in energy data alone. They also account for almost all of the apparent accuracy disagreement between stacks: excluding them takes the cross-ecosystem spread on VGG-16 on CIFAR-100 from 20.2 to 1.3 percentage points. A design with one run per configuration reports whichever of the two outcomes its single seed produced.

The two instruments agree on *energy* to 0.5% over 12600 blocks, which relocates the difficulty with software-based estimation. What does not agree is time: the duration CodeCarbon reports beside each energy figure is not the interval the energy was accumulated over but the phase plus a constant 3.28s, for every phase shorter than about 11s ($R^2 = 0.9980$; the excess is bimodal, either that constant or nothing). Power derived by dividing the one field by the other is therefore understated by up to 6.7 for sub-second phases and is exact above ten seconds — a bias that lands squarely on the fastest ecosystems and on the inference phase, which is to say on precisely the comparisons such a study exists to make.

The contributions of this work are as follows:

- A systematic empirical comparison of DL energy consumption across 7 language–framework ecosystems for GPU-based workloads, replicated 5 times per configuration and reported with between-run uncertainty and quality normalisation;
- An experiment specification for cross-ecosystem energy studies, together with an executable conformance checker that enforces it, and a single measurement contract implemented by all 7 stacks;
- A dual-instrument characterisation of CodeCarbon against hardware energy counters over 12600 measurement blocks, including a quantified failure mode of its reported duration that invalidates energy–time analyses built on it;

- A comprehensive replication package, including all data, scripts, and settings required to fully reproduce our results.¹

Improving the energy efficiency of AI systems is not just a technical challenge but also an ethical imperative as the field matures and scales [29]. In this regard, providing concrete empirical evidence on the energy consumption of different language–framework ecosystems represents a crucial first step towards the development of AI that is not only powerful, but also truly sustainable.

2. Background

The notion of *Green AI* [12, 13, 17] promotes the development of AI systems that are not only accurate but also energy-efficient and sustainable. Several indicators can be used to quantify sustainability, such as carbon footprint, financial cost, or energy usage [7, 5]. In this work we focus on **energy consumption**, measured in Joules, as it represents a direct and reproducible measure of computational effort. Unlike carbon footprint, which is heavily influenced by the energy mix and geographic location of execution, energy consumption can be compared more consistently across heterogeneous DL stacks under a uniform measurement protocol.

We treat execution time as a secondary metric rather than a proxy for energy. The prevailing position in the software-energy literature is that lower runtime does not imply lower energy [28, 30, 18], and we test it directly in Section 5.4 rather than assuming it. Our result differs from that literature, and the difference is instructive: on counter-bracketed durations, energy and time rank our configurations almost identically. Those studies largely concern CPU-bound general-purpose benchmarks, where a language implementation can trade instruction count against memory traffic and shift energy per second substantially. In a GPU-bound DL workload the accelerator dominates and holds a narrow band of power, so time and energy are close to proportional. We report this as a scope condition on the established finding, not a refutation of it — and we show in Section 5.5 that a study of exactly our shape can obtain either answer depending on which field of the instrument’s output it reads as the duration.

Validating the instrument is itself an active line of work. Comparative studies run software estimators alongside hardware wattmeters and report close tracking under controlled conditions and larger deviations on heterogeneous infrastructure [31, 32]; independent validation against external ground truth across hundreds of AI experiments reports estimator errors up to 40% and systematic underestimation of 20–30% in some settings [33]. That work asks whether an estimator’s number is close to the truth.

¹Replication package available at: <https://zenodo.org/records/17734884>

We ask a different and complementary question: whether a study can *tell* when it is not. The distinction matters because the two lead to different remedies. If the problem is accuracy, the remedy is a better estimator or a correction factor. If the problem is that an estimator silently substitutes a model for a measurement, reports a floor in place of a duration, or degrades below its own sampling interval, then no correction factor helps: what is needed is a second instrument over the same window, so that agreement becomes an observable property of each block. Our findings in Section 5.5 suggest the second framing is the more useful one for cross-ecosystem work, since on this platform the estimator’s *energy* turned out to be accurate and its *duration* turned out not to be a duration.

Deep Learning workloads are typically divided into two distinct phases: training and inference. *Training* requires iterative updates of model parameters through backpropagation and is generally the most computationally expensive stage. *Inference*, by contrast, involves only forward passes on a trained model and is usually less resource demanding. Both phases, however, display different energy profiles—*i.e.* optimizations in one phase do not necessarily carry over to the other [14, 15]. This motivates the need to analyze the two phases separately when evaluating their energy efficiency.

Language–framework ecosystems may influence the energy efficiency of DL workloads through a combination of factors, including framework execution engines, runtime overheads, data input pipelines, numerical precision policies, and the underlying driver/toolchain configuration. While compiled and interpreted languages differ in their execution models, in GPU-based DL workloads much of the computation is typically delegated to optimized backend kernels; therefore, differences observed in practice should be interpreted at the *ecosystem level* rather than attributed to the programming language alone. Moreover, some ecosystems (e.g. MATLAB with its Deep Learning Toolbox, or Java with Deeplearning4j) introduce additional runtime layers compared to lower-level bindings (e.g. LibTorch in C++ or `torch` in R), which can affect both performance and energy behavior. Understanding these ecosystem-level differences is crucial, since language choice is often treated as a secondary engineering decision, yet in practice it constrains the set of available frameworks, runtimes, and supported toolchains, ultimately affecting the energy profile of DL model training and inference.

Energy measurement for workloads of this kind can be read at three places, and the choice of place matters more than the choice of tool. A *wall meter* sees the whole machine including power supply losses, fans and drives. *Hardware energy counters* — NVML’s `nvmlDeviceGetTotalEnergyConsumption` on NVIDIA

accelerators and Intel RAPL on the CPU package — accumulate energy in the silicon itself and are read by software but not estimated by it. *Software estimators* such as **CodeCarbon** [2] sit on top of whatever the platform exposes and substitute a model wherever it exposes nothing.

Software estimators have become standard in sustainability-oriented AI research [29, 8], and their reliability for ML and DL workloads is documented across model compression analysis [34], optimizer efficiency benchmarks [35], large-scale evaluations of Large Language Models [36] and cybersecurity systems [37]. Comparative studies have also run **CodeCarbon** alongside hardware wattmeters in the same setup and report that software estimates track sensor readings closely under controlled conditions while deviating more on heterogeneous infrastructure [31, 32]. Independent validation against external ground truth across hundreds of AI experiments nonetheless reports estimator errors of up to 40%, and systematic underestimation of 20–30% in some settings [33].

We take a different position from the one this literature usually implies. The question is not whether a software estimator is accurate enough to be trusted, but whether a study can tell when it is not. An estimator that falls back on a model when a counter is unavailable, and announces the substitution only in a log line, produces a number of the same shape and plausibility either way. We therefore instrument every measurement block *twice*: hardware counters as the primary instrument, and **CodeCarbon** over the identical window as an independent second reading. Agreement between the two is then an observable property of each block rather than an assumption, and the places where they cannot agree — CodeCarbon’s RAM term is derived from installed memory capacity, and no counter on this platform can confirm or refute it — are identified rather than absorbed.

This also fixes the boundary. NVML plus RAPL is a *chip-level* measurement: it excludes the power supply, fans and drives that a wall meter would see, and must not be presented as a whole-system figure. We had no wall meter available and say so, reporting a chip-boundary quantity consistently rather than a mixture of measured and modelled terms that corresponds to no physical boundary at all. Section 5.5 reports what the two instruments say about each other across the campaign.

3. Research Methodology

This work adopts an empirical methodology grounded in the principles of software engineering empirical experimentation [38, 39, 40, 41]. Our objective is not to propose a new algorithm or framework, but to

systematically analyze how choosing among language–framework ecosystems (including their supported runtime and toolchain) can impact the energy efficiency of DL workloads. To this end, we designed an *in vitro* controlled experiment that treats ecosystems as independent variables, DL workloads as experimental objects, and energy consumption as the primary dependent variable [42, 43, 44].

Research goal and questions. Following the Goal-Question-Metric (GQM) paradigm [42], our goal is formulated as:

*Analyze DL training and inference,
for the purpose of quantifying and compar-
ing their energy consumption,
with respect to the choice of DL ecosys-
tems (i.e. language bindings, framework-
s/libraries, supported runtimes/execution
engines, and associated toolchains),
from the viewpoint of software engineering
researchers and practitioners,
in the context of computer vision image
classification workloads.*

Research questions. This study investigates how language–framework ecosystem selection influences the energy efficiency of Deep Learning (DL) workloads. We focus on three key questions:

- **RQ1:** *How does language–framework ecosystem choice affect the energy efficiency of DL training?*
- **RQ2:** *How does language–framework ecosystem choice affect the energy efficiency of DL inference?*
- **RQ3:** *How do energy efficiency and execution time relate across different DL ecosystems?*
- **RQ4:** *To what extent are the answers to RQ1–RQ3 determined by the measurement apparatus rather than by the ecosystems?*

RQ4 is not a robustness check appended to the other three. A cross-ecosystem energy comparison is an inference from an instrument to a claim about software, and the inference is only as good as the guarantee that the instrument reads the same quantity, over the same window, for every stack. We therefore treat the apparatus as an object of measurement in its own right: every block is recorded by two independent instruments, and RQ4 asks what they say about each other and about the answers to RQ1–RQ3.

Objects of study. We focus on convolutional neural networks (CNNs), which remain fundamental in modern computer vision tasks [4, 14, 15]. Specifically, we consider two canonical architectures: ResNet-18 [45] and VGG-16 [46]. These models are selected not only for their distinct computational characteristics (residual connections vs. sequential depth), but

primarily because they are natively provided *off-the-shelf* by the vast majority of DL frameworks. By relying on these standard, pre-implemented architectures, we ensure that our measurements capture the frameworks’ internal optimizations and native performance, rather than introducing researcher-specific biases that would result from implementing the models from scratch.

To account for varying workload complexity, we employ three widely utilized datasets. *Fashion-MNIST* provides a lightweight baseline with 28 28 grayscale images across 10 classes [47], and is often used as a modern drop-in replacement for MNIST². It includes 60,000 training and 10,000 test images. *CIFAR-100* increases complexity with 32 32 color images across 100 categories, remaining a canonical benchmark for image classification [48], and comprises 50,000 training and 10,000 test images. *Tiny ImageNet* represents a demanding workload with 64 64 color images across 200 classes and 100,000 training and 10,000 test images, serving as a proxy for ImageNet-scale tasks [10].

All datasets were resized to a uniform resolution of 32 32 pixels. This preserves the relative differences in complexity (grayscale vs. color, number of classes, dataset size) while removing input resolution as a confounding factor [27, 26]. In addition, standardizing input size avoids the need to modify the final classification layer of the networks for each dataset, thus keeping model architectures fixed across experiments. This design choice not only improves comparability of results but also reflects realistic developer practice, where pre-trained or canonical models are often reused with minimal architectural modifications [17, 13]. Finally, a common resolution improves stability across frameworks, some of which (e.g. Burn for Rust, Deeplearning4j for Java) offer more reliable performance with standardized image sizes [18, 21]. Nevertheless, fixing a 32 32 input resolution may interact with framework-specific memory and kernel selection strategies; therefore, conclusions should not be extrapolated to higher-resolution workloads without additional experiments.

Independent and dependent variables. Our primary independent variable is the *DL ecosystem*, i.e. the language–framework stack as adopted in practice (including its officially supported runtime and CUDA/cuDNN toolchain). In our design, each ecosystem is treated as a single categorical treatment; we do not attempt to decompose effects into separate contributions of language, framework, runtime, or toolchain. Specifically, we evaluate 7 language-based ecosystems: Python (PyTorch, TensorFlow, JAX), C++ (LibTorch), Java (Deeplearning4j), R (torch), and Rust (tch). A second independent variable is the *DL phase* (training and inference).

MATLAB with the Deep Learning Toolbox was part of an earlier campaign and is out of scope here. The reason is specific rather than dismissive: the toolbox

²<https://github.com/zalandoresearch/fashion-mnist>

does not expose the training loop at the granularity the specification of Section 3.1 requires, so its data pipeline, its evaluation batching and its measurement boundaries cannot be brought into conformance with the other 7. Including a stack that provably runs a different experiment would reintroduce exactly the confound this design exists to remove.

Our dependent variables are the *energy consumption* of each measurement block, in Joules at a chip-level boundary (accelerator plus CPU package), and the *test accuracy* reached at the end of each epoch. Recording the second alongside the first is what allows energy to be normalised by achieved quality in Section 5.2; a fixed-budget comparison that does not record accuracy cannot distinguish an efficient stack from one that has simply learned less [5, 29, 6].

Study design. We adopt a *black-box design*: models are executed using off-the-shelf implementations under identical hyperparameters (e.g. learning rate, batch size) [20, 19, 30]. This choice enforces the realism of the study by reflecting how practitioners use frameworks *in vivo*, rather than relying on ad-hoc or reimplemented versions.

By keeping hyperparameters constant, we implement a controlled *fixed-budget* comparison across ecosystems (i.e. identical training recipe and number of epochs). This design ensures a fair baseline comparison in which we minimize the risk of introducing researcher-specific implementation biases, thereby improving comparability; however, it does not guarantee convergence equivalence across frameworks. This represents an explicit trade-off: allowing per-ecosystem hyperparameter tuning could yield better absolute efficiency for each stack, but would confound whether differences stem from the ecosystem itself or from the tuning process.

Moreover, we treat framework-level execution engines and optimizations (e.g. graph compilation, runtime schedulers, backend kernels) as *part of the ecosystem* rather than normalizing them away. As a consequence, observed differences should be interpreted as ecosystem-level effects, not as isolated language-level behavior.

Numerical precision policy. We retained the default numerical precision and mixed-precision policies of each ecosystem (e.g. FP32 vs mixed precision), treating them as part of the ecosystem-level behavior observed in practice. Consequently, differences in precision paths may contribute to measured energy and execution time; we therefore avoid attributing observed effects to the programming language alone.

3.1. What “the same experiment” means, and how it is enforced

A fixed-budget comparison presumes that the budget is the same. In practice each ecosystem is written by a different person against a different API, and the places where they can silently diverge are numerous,

individually reasonable and invisible in the energy output: a default learning rate that differs between two framework APIs, a data loader whose worker count follows the machine rather than the protocol, an evaluation loop written one image at a time because the API made that natural, an input tensor left at its native resolution in one stack and resized in another.

We therefore wrote the protocol down before running anything, as a specification with six parts, and made conformance machine-checkable.

- **S1 — Model.** The four ecosystems that share the LibTorch backend (Python, C++, R, Rust) train the *same* TorchScript module, exported once per (architecture, dataset) pair, rather than four independent reimplementations of ResNet-18 and VGG-16. The remaining stacks use the canonical architecture their framework ships, with parameter counts checked against the exported module.
- **S2 — Optimisation.** One optimiser, one learning rate, one batch size, one epoch count, one loss, one seeding policy, applied identically and asserted at startup.
- **S3 — Pipeline.** One input specification (32 32 3, raw [0,1] inputs, no per-stack normalisation), a bounded and equal number of loader workers, and batched evaluation at the training batch size in every stack.
- **S4 — Backend.** The accelerator is asserted at startup and CPU fallback is refused rather than warned about. The LibTorch version is pinned across the four stacks that share it.
- **S5 — Measurement.** One measurement bridge, one sampling interval, one tracking mode, one set of counters, one unit, and per-epoch accuracy persisted alongside energy.
- **S6 — Replication.** Each configuration is executed 5 times as independent processes with distinct seeds, interleaved rather than run back to back.

The specification is enforced by 57 automated checks that read the source of every ecosystem and the configuration of every tracker, and fail the build when a stack drifts out of conformance. This matters more than it may appear. A specification that lives only in a paper is a description of intent; the same specification expressed as an executable check is a property of the artefact, and it fails at the moment of the divergence rather than at peer review. Two of the defects we corrected during development were introduced by us and caught by these checks rather than by inspection.

Residual, documented divergences. Two could not be removed and are reported rather than hidden.

DeepLearningJ 1.0.0-M2.1 is the last release of its API and links against the CUDA 11.6 runtime, so the Java stack cannot be brought onto the CUDA 12 toolchain the other six share. And the R `torch` binding cannot switch a script module between training and evaluation mode, so R alone cannot consume the shared TorchScript module of S1 and builds an equivalent architecture instead. Both are properties of the ecosystems, and arguably part of what one measures when one measures an ecosystem, but they are not the binding overhead the study sets out to isolate.

3.2. Measurement procedure

Energy is read from hardware counters at the boundaries of each measurement block: NVML’s `nvmlDeviceGetTotalEnergyConsumption`, an accumulating millijoule register on the accelerator, and the RAPL package-energy counters exposed at `/sys/class/powercap` for the CPU, with wraparound at `max_energy_range_uj` handled explicitly. The reported quantity is their sum: a chip-level boundary, identical for every ecosystem.

CodeCarbon [2] runs over the same window as an independent second instrument, in machine tracking mode at a one-second sampling interval, configured identically for all 7 stacks through a single shared bridge. The six non-Python ecosystems drive that bridge over a line protocol with synchronous acknowledgement, so that a phase boundary in the measured process and the corresponding counter read cannot drift apart — an earlier asynchronous version of the same bridge under-reported one stack’s training energy by 24% and recorded its evaluation energy as zero.

A measurement block is one phase of one epoch: 12600 blocks in total, each carrying two energy readings, a duration, and the test accuracy reached. Because tracking is machine-wide, no other work runs on the host during a campaign; blocks measured while a background download was in progress during development were discarded and the protocol amended rather than corrected after the fact.

Replication and the unit of analysis. Each of the 42 configurations is executed 5 times as an independent process with a distinct seed, giving 210 runs. The repetitions are *interleaved* rather than run consecutively, so that drift in machine state — thermal, background, driver clock behaviour — is spread across conditions instead of aliasing onto whichever ecosystem happened to run last. All uncertainty we report is between-run: the 30 epochs within a run share an initialisation, a JIT outcome, an allocator state and a thermal trajectory, and treating them as repeated measurements would understate uncertainty while inflating apparent significance.

Quality normalisation. Because every stack records test accuracy per epoch, we report energy in three forms: energy per epoch, energy per unit of accuracy

achieved, and the energy required to first reach a target accuracy fixed per dataset. The last is the closest to a practitioner’s question, and is the one a fixed-budget design cannot answer on its own.

Reproducibility and rigor. We fixed pseudo-random seeds for data shuffling and model initialization where supported by the ecosystem. Nevertheless, full determinism cannot be guaranteed across all stacks due to non-deterministic GPU kernels and differences in backend execution engines; this is one reason the design replicates rather than relying on a single execution per configuration.

A complete replication package is provided (see replication package link in Section 1), including: (i) source code for all implementations, (ii) scripts to reproduce total (iii) ecosystem specifications (e.g. virtual environments, Maven, CMake), and (iv) collected raw data [17, 13]. Configurations are standardized across languages where feasible, with deviations documented when framework-specific requirements arise [18, 21]. This package enables replication, validation, and extension under different hardware or software conditions [38, 40].

4. Experimental Execution

This section describes the practical execution of the study, complementing the research design presented in Section 3. Here we document the experimental setup, implementation details, measurement procedure, and reproducibility package, following a step-by-step process to ensure transparency [38, 41].

Experimental setup. All experiments reported here were executed on a single dedicated workstation: an NVIDIA GeForce RTX 3090 (Ampere, 24 GB GDDR6X, 350 W board power limit, driver 595.84), an AMD Ryzen 9 5900X (12 cores / 24 threads), 30 GB of RAM and NVMe storage, running Ubuntu 26.04 LTS (kernel 7.0.0). The machine was otherwise idle for the duration of the campaign, which whole-machine energy tracking requires.

Two properties of this platform bear directly on the measurements. First, both energy counters are available: the RTX 3090 exposes NVML’s accumulated-energy register, and the Ryzen’s package-energy counters are readable through the Linux `powercap` interface — which, for historical reasons, presents AMD domains under `intel-rapl` names. We read the package domain and handle counter wraparound at `max_energy_range_uj` explicitly. Second, this is a desktop rather than a server: it has one accelerator, one CPU socket and an order of magnitude less RAM than a datacentre node, which is precisely why the modelled RAM term discussed in Section 5.5 is a smaller share of a software estimator’s total here than it would be on a large-memory host.

We report this as a replication platform rather than as the platform. An earlier campaign of the same design

ran on a dual-socket Intel Xeon Gold 5418Y server with an NVIDIA L40S; absolute energy figures are not comparable between the two machines, and none of the comparisons in this paper is made across them. What carries over is the design, the specification and the instrument characterisation, all of which are properties of the method rather than of the host.

Model implementations. Every ecosystem uses the canonical architecture its framework provides rather than a reimplement, and, where the backend permits, the *same* artefact. Under specification S1 the four LibTorch-based stacks — Python, C++, R and Rust — do not build four independent ResNet-18s and VGG-16s but load one TorchScript module per (architecture, dataset) pair, exported once and version-pinned in a manifest that the conformance checker compares against the LibTorch each stack links. This removes what would otherwise be the largest uncontrolled variable in the study: in an earlier campaign these four stacks differed in parameter count, and comparing them was therefore comparing four models, not four bindings.

Table 1 lists the versions. The Python stacks are installed in three separate virtual environments rather than one, because their CUDA and cuDNN wheels conflict: a shared environment resolved to a cuDNN that TensorFlow could not dlopen, and TensorFlow then ran unaccelerated while reporting only a warning — a failure mode the accelerator assertion of S4 now turns into an error.

Toolchain divergence, and what is left of it. The four LibTorch stacks are pinned to one LibTorch build (2.7.0, CUDA 12.8), so within that control group the backend is genuinely held constant. Two ecosystems cannot join it. DeepLearning4j 1.0.0-M2.1 links against CUDA 11.6 and cuDNN 8.3 and has had no subsequent release; the stack now carries its own redistributables through `org.bytedeco:cuda-platform-redist`, which makes it reproducible without making it contemporary. R’s `torch` 0.17.0 bundles its own LibTorch and resolves CUDA at load time. We report these as documented divergences rather than normalising them away, and Section 5.2 reports the LibTorch control group separately so that a reader can see how much of the total spread survives when the backend is held fixed.

During a preliminary pilot phase, the Julia programming language (using `Flux` and `Lux`) was also considered. However, persistent technical challenges encountered during implementation related to gradient updates during training, which remained unresolved also after directly contacting the developers of the libraries, prevented its reliable inclusion in the final study.

Procedure. As summarized in Table 2, each experimental configuration (model dataset ecosystem) is executed with consistent hyperparameters tuned to

balance comparability and realism [4, 14]. Training runs are performed for 30 epochs with a batch size of 128, using the Adam optimizer with a fixed learning rate of $1 \cdot 10^{-4}$. Loss is computed via cross-entropy, consistent across all ecosystems. Input preprocessing includes resizing all images to 32 32 pixels, with grayscale datasets (e.g. Fashion-MNIST) expanded to three channels for compatibility with CNN backbones. These settings reflect canonical practices in the community while ensuring comparability across implementations [10, 18].

Measurement. Each phase of each epoch is bracketed by two counter reads — NVML’s accumulated-energy register and the CPU package RAPL counter — and is simultaneously tracked by `CodeCarbon` v3.3.0 [2] over the identical window. The two instruments are started and stopped by the same code path, so their windows cannot drift apart.

The six non-Python ecosystems drive the instrumentation through a single shared bridge: a Python process that accepts `START`, `STOP`, `METRIC` and `EXIT` over a line protocol, reads the counters, and acknowledges each command before the measured process proceeds. The acknowledgement is not incidental. An earlier, asynchronous version of the same bridge allowed a phase boundary in the measured process to run ahead of the corresponding counter read, which under-reported one stack’s training energy by 24% and recorded its evaluation energy as exactly zero — a value that looks like a bug only if someone thinks to look at it.

Running the bridge out of process also isolates the measured stack from the instrument: a crash in the tracker cannot take down a six-hour training run, and the tracker’s own CPU cost is attributed to the machine total rather than silently to the workload. All runs execute in isolation on an otherwise idle host, as whole-machine tracking requires, and each block is logged with dataset, model, ecosystem, repetition, seed, phase, epoch, both energy readings, the duration and the test accuracy reached.

Reproducibility. A comprehensive replication package accompanies this study (see the link in Section 1). Beyond source code, automation scripts, per-ecosystem environment specifications and raw logs, it contains three artefacts specific to this design:

- (i) the **experiment specification** of Section 3.1, written before the campaign and versioned with it;
- (ii) the **conformance checker**, 56 executable checks that read the source of every ecosystem and the configuration of every tracker, and fail when a stack drifts out of specification;
- (iii) the **dual-instrument records**: every block’s counter reading and estimator reading side by side, which is what makes the analysis of Section 5.5 reproducible rather than anecdotal.

Table 1

Ecosystems, frameworks, versions and backends. The four stacks marked £ share one pinned LibTorch build and one exported TorchScript module, and form the internal control group of the study.

Ecosystem	Framework / Library	Version	Backend
Python	PyTorch / TorchVision£	v2.7.0 / v0.22.0	LibTorch 2.7.0, CUDA 12.8
	TensorFlow / Keras	v2.19.1 / v3.15.1	cuDNN 9.3, CUDA 12 wheels
	JAX / Flax / Optax	v0.10.2 / v0.12.8 / v0.2.8	cuDNN 9.24, CUDA 12 wheels
C++	LibTorch£	v2.7.0	LibTorch 2.7.0, CUDA 12.8
Java	Deeplearning4j	v1.0.0-M2.1	ND4J CUDA 11.6, cuDNN 8.3
R	torch£	v0.17.0	bundled LibTorch, CUDA 12
Rust	tch£	v0.20.0	LibTorch 2.7.0, CUDA 12.8
<i>Instrumentation</i>			
	CodeCarbon	v3.3.0	identical config, all stacks
	NVML / RAPL counters	driver 595.84	read by the shared harness

Table 2

Hyperparameter configuration, identical across all ecosystems and enforced by the conformance checks of specification S2 rather than by convention.

Parameter	Value
Number of epochs	30
Batch size	128
Optimizer	Adam
Learning rate	1 · 10 ⁻⁴
Loss function	Cross-entropy
Input resolution	32 32 (all datasets)
Input channels	3 (grayscale datasets expanded)
Accelerator	asserted at startup; CPU fallback refused
Data loader workers	2, bounded and equal in every stack
Input normalisation	none; raw [0, 1] inputs everywhere
Evaluation	batched at the training batch size
Independent runs	5 per configuration, interleaved

Every number in this manuscript is generated from those records by the analysis pipeline and injected into the source at build time; none is transcribed.

5. Results

We report the campaign in the order the research questions were posed, with one departure: RQ4, on the apparatus, comes last but conditions everything before it, and a reader who wants to know how much to trust Sections 5.2–5.4 should read Section 5.5 first.

All figures below are between-run. Each configuration contributes 5 independent runs, and every interval, test and effect size is computed across those runs rather than across the epochs within one.

5.1. Repeatability of the measurements

Before comparing ecosystems it is worth asking how precisely the apparatus measures any single one of them, because the answer bounds every comparison that follows and because the submitted design could not produce it at all.

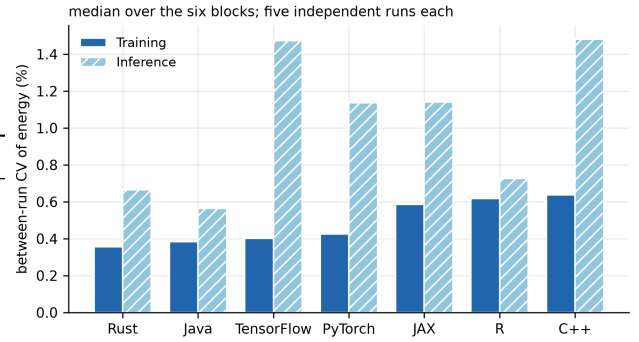


Figure 1: Between-run coefficient of variation of energy, median over the six (model, dataset) blocks, from 5 independent runs per configuration.

Across all 42 configurations, the median between-run coefficient of variation of training energy is 0.44 % and the worst is 1.29 %. Inference, whose blocks are one to two orders of magnitude shorter, is noisier but still tight: a median of 1.06 % and a worst case of 9.16 %. Figure 1 breaks this down by ecosystem.

Two things follow. First, differences of a few percent between ecosystems are resolvable, and the differences we report below are far larger than that. Second, the run-to-run variability is small enough that a single run per configuration would have produced a plausible-looking number — which is precisely why a single run per configuration is not evidence that the number is right. Low variance is a property of the measurement, not a guarantee about the thing measured.

5.2. RQ1: Impact of ecosystems on training energy efficiency

Figure 2 and Table 3 report training energy per epoch for every ecosystem and block, at a common chip-level boundary.

The spread between the cheapest and the most expensive ecosystem ranges from 7.8 to 19.1 depending on the block (Table 4), with JAX cheapest in every one. This is a substantially larger effect than the 4.6 an epoch-averaged, single-run, mixed-boundary analysis

GPU + CPU package, one boundary for every ecosystem; bars are 95% between-run intervals over five independent runs

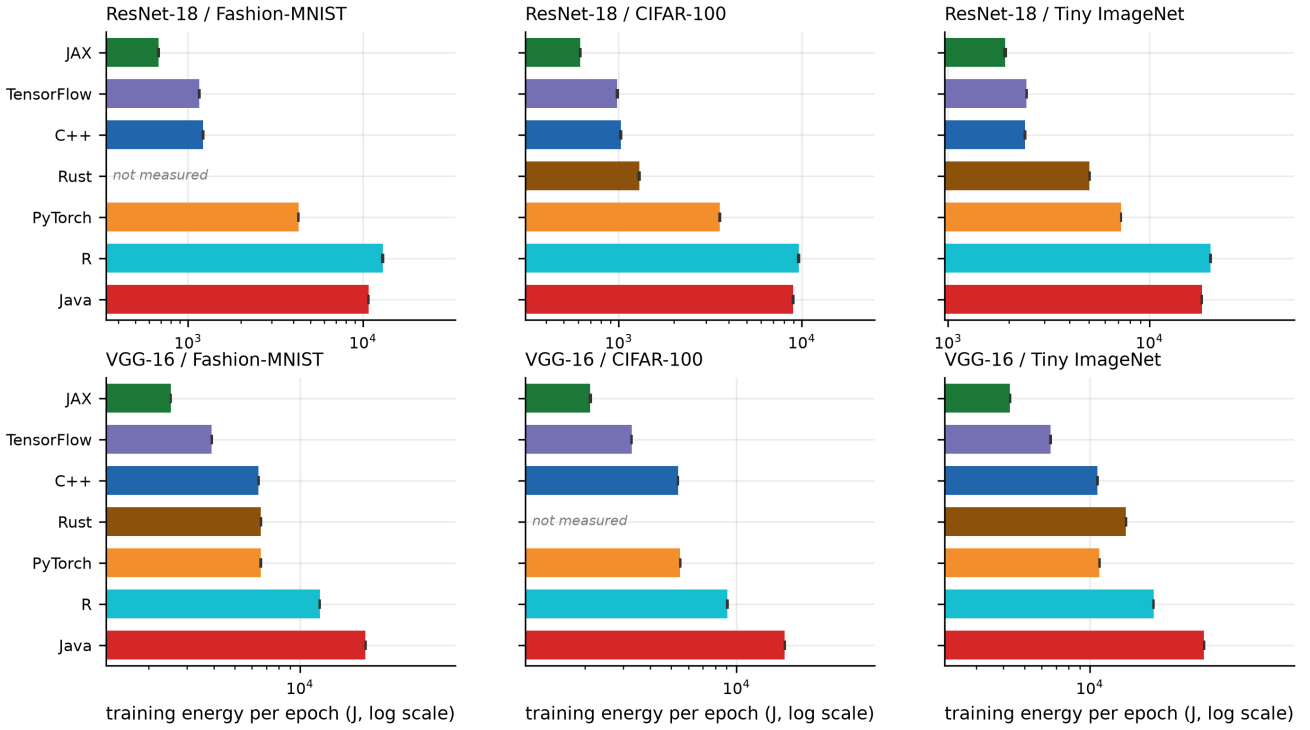


Figure 2: Training energy per epoch, accelerator plus CPU package, with 95% between-run intervals. Note the logarithmic axis: the intervals are narrow enough to be hard to see at this scale.

Table 3

Mean training energy per epoch (J), accelerator plus CPU package.

Ecosystem	ResNet-18 / CIFAR-100	ResNet-18 / Fashion-MNIST	ResNet-18 / Tiny ImageNet	VGG-16 / CIFAR-100	VGG-16 / Fashion-MNIST
JAX	617	678	1 921	2 115	2 535
TensorFlow	982	1 160	2 454	3 283	3 569
C++	1 025	1 220	2 408	5 369	5 498
Rust	1 294	—	5 028	—	—
PyTorch	3 565	4 288	7 217	5 498	5 498
Java	8 980	10 770	18 164	16 719	16 719
R	9 619	12 967	20 053	9 065	9 065

of a campaign of this shape would report, and it is larger for a straightforward reason: once every stack is measured at the same boundary, the constant host term that used to be added to all of them — compressing their ratios toward one — is gone.

The shared-backend control group. Four of the 7 ecosystems run on LibTorch, and three of those four — Python, C++ and Rust — train the byte-identical TorchScript module under specification S1. Whatever separates them cannot be the kernels, because the kernels are the same code; it is binding overhead, data movement and host-side dispatch. Table 5 reports the spread within that group against the spread across all 7. A large control-group spread would mean the effect is host-side rather than intrinsic to the framework; a small one means the framework choice genuinely dominates.

Table 4

Cheapest and most expensive ecosystem per block, training energy per epoch.

Model	Dataset	Lowest		Highest	
		J		J	Ratio
ResNet-18	CIFAR-100	JAX	617	R	9 619 (15.6)
ResNet-18	Fashion-MNIST	JAX	678	R	12 967 (19.1)
ResNet-18	Tiny ImageNet	JAX	1 921	R	20 053 (10.4)
VGG-16	CIFAR-100	JAX	2 115	Java	16 719 (7.9)
VGG-16	Fashion-MNIST	JAX	2 535	Java	20 094 (7.9)
VGG-16	Tiny ImageNet	JAX	4 287	Java	33 534 (7.8)

Energy at equal quality. A fixed-budget comparison rewards a stack for learning less. Because every ecosystem now records test accuracy per epoch, we can check whether it is doing so. Table 6 reports final

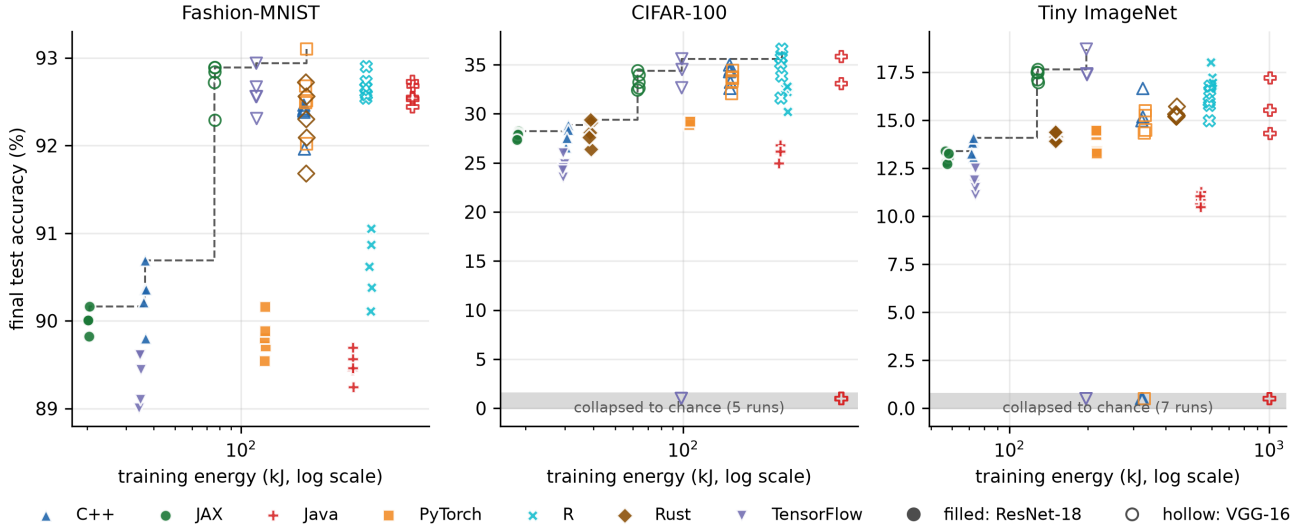


Figure 3: Energy spent against accuracy reached, one point per run. The dashed line is the empirical frontier: no configuration to its left reaches the same accuracy for less energy.

Table 5

Spread within the shared-backend control group against the full spread, per block. The control group holds LibTorch, the model and the data pipeline fixed.

Model	Dataset	All 7	LibTorch group	Share of log spread
ResNet-18	CIFAR-100	14.9	9.6	84%
ResNet-18	Fashion-MNIST	18.3	10.8	82%
ResNet-18	Tiny ImageNet	10.2	8.5	92%
VGG-16	CIFAR-100	7.9	1.7	25%
VGG-16	Fashion-MNIST	7.9	1.9	32%
VGG-16	Tiny ImageNet	7.8	1.8	29%

Table 6

Final test accuracy (%) after the fixed epoch budget, mean over 5 runs.

Ecosystem	Fashion-MNIST	CIFAR-100	Tiny ImageNet
C++	91.30	30.98	11.60
Java	91.03	20.27	10.23
JAX	91.36	30.56	15.19
PyTorch	91.19	31.23	12.99
TensorFlow	90.98	22.79	11.31
R	91.64	33.34	16.57
Rust	92.27	28.10	14.74

test accuracy by ecosystem and dataset. On Fashion-MNIST all 7 stacks land within 1.3 percentage points of one another (91.0–92.3%), which is the strongest available evidence that they are running the same experiment — and a check the submitted design could not perform, because no ecosystem wrote its accuracy to disk. On the harder datasets the stacks separate more (CIFAR-100: 20.3–33.3%), and there the energy comparison must be read alongside the accuracy reached rather than instead of it.

Figure 3 plots the two together. Ranking ecosystems by accuracy per kilojoule rather than by energy alone puts JAX ahead of Java by 13.7 — a different quantity

from the energy ratio, and the one a practitioner choosing a stack actually faces.

5.2.1. Runs that consume the budget and learn nothing

Replicating each configuration surfaced something a single run cannot show. VGG-16 does not always train. In 12 of 100 VGG-16 runs the network converges to *exactly* chance accuracy — 1/100 on CIFAR-100, 1/200 on Tiny ImageNet — and stays there for all 30 epochs, having consumed the full energy budget while learning nothing. ResNet-18 never does this: 0 of 100 runs. Neither architecture does it on Fashion-MNIST.

The pattern points at the recipe rather than at any ecosystem. VGG-16 as shipped carries no batch normalisation, and a plain sixteen-layer network initialised randomly and driven by Adam at 10^{-4} over a hundred or two hundred classes can settle into predicting a single class; whether it does is decided by the initialisation. The collapse appears in 4 ecosystems independently, and 3 never exhibit it. Per-ecosystem rates range from 0% to 50% against an overall 18%, but with ten susceptible runs per ecosystem that spread is not distinguishable from chance (permutation test, $p = 0.12$). We therefore report the susceptibility as a property of the recipe and leave open whether ecosystems differ in rate — they plausibly do, since each framework initialises weights with its own scheme and its own generator, but resolving it needs more repetitions than an energy study can usually afford.

A failed recipe and a broken pipeline look identical in an energy table. Both produce chance accuracy at full energy cost, and neither is visible in a Joule figure. The per-epoch traces separate them completely. In a genuine collapse the training loss does not move either — across all 12 of them it changes by at most 0.0%, because a network that has settled on

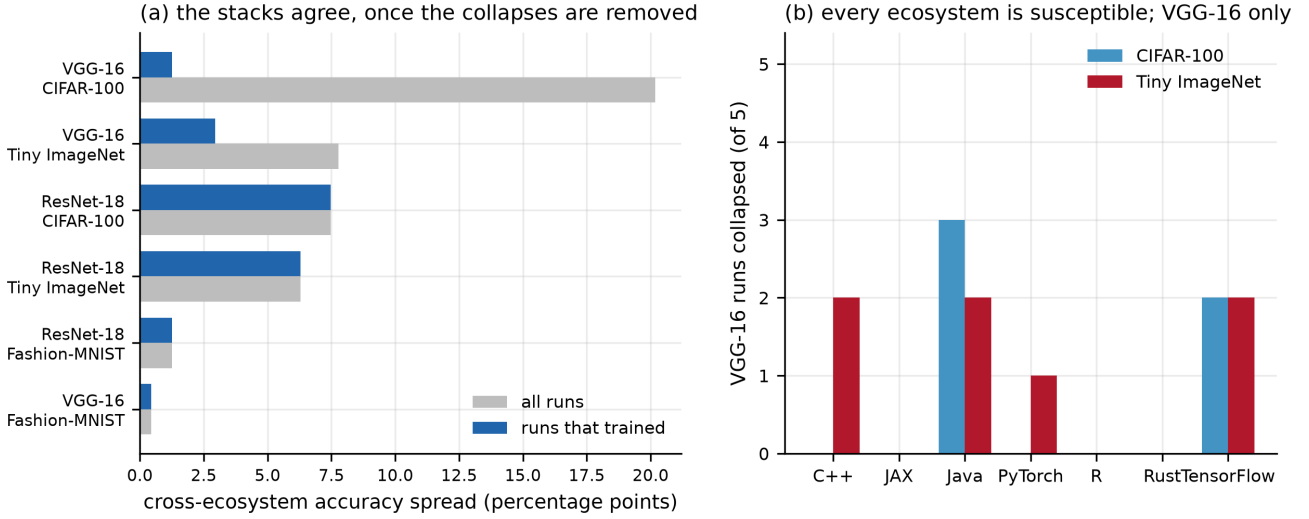


Figure 4: (a) Cross-ecosystem accuracy spread per block, over all runs and over only the runs that trained. (b) VGG-16 runs that collapsed to chance, out of five per cell.

one class has nothing left to fit. A pipeline defect looks the opposite: the training loss falls normally while test loss *rises* and accuracy stays at chance, which says that training and evaluation are not seeing the same inputs.

We report this because the discriminator found a defect of our own that the collapse count alone would have absorbed. 4 runs met the chance-accuracy criterion — at most 14.7% — with training loss falling by 86–87%: one Rust binary carried a private copy of the input transform, applied on the training path only, which resized the tensor its loader had already produced using a function that returns `uint8`. Every value in $[0,1]$ truncated to zero, so it trained on black images and was evaluated on real ones. Filed as collapses they would have read as “VGG-16 sometimes fails to train”.

They were a defect, it is fixed, and the configuration was re-executed: those runs now reach the same accuracy as every other stack, and the discriminator finds 0 pipeline defects in the campaign as it stands. The 12 collapses reported above are the ones that survive the test. Since the evidence for a defect should be as traceable as the evidence for a result, and the discarded runs no longer exist in the campaign, their signature is preserved as a record in the replication package rather than quoted from memory.

Two consequences matter here.

It is the strongest evidence that the specification worked. On VGG-16 on CIFAR-100 the cross-ecosystem accuracy spread is 20.2 percentage points over all runs and 1.3 points over the runs that trained (Figure 4a). Almost the entire apparent disagreement between stacks was a handful of runs that failed to train, not a difference in what the stacks compute. No block exceeds 7.5 points once collapses are excluded.

It breaks fixed-budget energy comparison, invisibly. A collapsed run costs full energy and produces

nothing: 5.1 MJ, or 10.7% of the campaign’s total, was spent on runs that learned nothing. An ecosystem that drew unlucky initialisations looks equally expensive and much less accurate; one that drew lucky ones looks efficient. With a single run per configuration — the design this study replaces — the reported number is whichever outcome that one initialisation produced, and nothing in the energy data distinguishes the two cases. This is why we report accuracy alongside energy rather than assuming a fixed budget delivers fixed quality.

RQ1 Findings: Impact of Ecosystems on Training Energy Efficiency

Measured at a common chip-level boundary and replicated 5 times per configuration, training energy differs across ecosystems by 7.8 to 19.1 depending on architecture and dataset, with JAX cheapest in every block. Between-run variability is small (median CV 0.44%), so these differences are well resolved. Final accuracy converges to within 1.3 percentage points on Fashion-MNIST, so the differences are energetic rather than qualitative — the stacks are doing the same work at very different cost.

5.3. RQ2: Impact of ecosystems on inference energy efficiency

Table 7 reports inference energy per epoch over the test set, at the same boundary.

The inference spread is wider than the training spread (11.0 to 53.0), and this is where the apparatus matters most. An inference pass at these dataset sizes takes well under a second in the fastest stacks, which places it below the reported-window floor characterised in Section 5.5. Any inference comparison drawn from an estimator’s own duration field is therefore comparing

Table 7

Mean inference energy per evaluation pass (J), accelerator plus CPU package.

Ecosystem	ResNet-18 / CIFAR-100	ResNet-18 / Fashion-MNIST	ResNet-18 / Tiny ImageNet	VGG-16 / CIFAR-100	VGG-16 / Fashion-MNIST
JAX	68	51	150	121	121
TensorFlow	73	58	156	136	136
C++	81	57	151	193	193
Rust	113	—	391	—	—
PyTorch	229	252	388	366	366
Java	588	584	616	1 150	1 150
R	2 411	2 676	2 672	2 349	2 349

floors rather than phases for exactly the stacks that are fastest. Our figures are counter-bracketed and do not have this problem, but the point generalises: the inference phase is where a cross-ecosystem energy study is most exposed to its instrument, and it is also the phase practitioners deploy.

RQ2 Findings: Impact of Ecosystems on Inference Energy Efficiency

Inference energy spreads more widely across ecosystems than training energy (11.0–53.0), and does so on blocks short enough that the measurement instrument, not the ecosystem, becomes the dominant source of error in a naive design. Reported inference rankings should be treated as claims about the apparatus until the window over which energy was accumulated is stated.

5.4. RQ3: Energy–time relationship across ecosystems

The claim that “faster is not greener” is a claim about two measurements, and it is only as good as the weaker of them. We therefore recompute it on counter-bracketed durations — the interval between the two counter reads that bracket a phase — rather than on any duration reported by the estimator.

On that basis, energy and time rank the configurations almost identically: Spearman $\rho = 0.98$ for training and 0.98 for inference, with 5.1% and 5.0% of configuration pairs discordant respectively. Within this workload and platform, execution time is a good proxy for energy, and the interesting question is why an analysis of the same experimental shape can conclude otherwise.

The answer is in Figure 5. Panel (a) plots the duration CodeCarbon reports for each block against the counter-bracketed phase, and the relation has two regimes. Above about 11s the two coincide to 15ms. Below it, the reported duration is the phase plus a constant 3.28s — the excess is bimodal, essentially either that constant or nothing, with no intermediate values. Modelling it that way accounts for $R^2 = 0.9980$ of the variance at a mean absolute error of 0.40s.

We report the fit this precisely because our own first attempt was a floor, $\max(\text{phase}, 3.99\text{s})$, which

reaches $R^2 = 0.9926$ and looks conclusive. It is not: that R^2 is carried by the very short and the very long blocks, where both models agree, and it is wrong in the middle, at nearly three times the error (1.13s). A high coefficient of determination on a skewed predictor is not evidence of the right functional form — which is the argument of this paper applied to our own analysis.

The consequence is unchanged either way. The shortest phase in the campaign takes 0.20s and is filed as 3.47s, a factor of 17.

Panel (b) shows the consequence. Power computed as reported energy over reported duration is understated by 6.7 for sub-second phases (40W against a measured 268W) and is accurate to 1% above ten seconds. The bias is not random and it is not uniform: it is a monotone function of phase length, so it falls hardest on the fastest ecosystems and on the inference phase, and it stretches their apparent time axis while leaving their energy untouched. That is precisely the shape of an artefactual “fast but energy-hungry” finding.

RQ3 Findings: Energy and Time

On counter-bracketed durations, energy and execution time rank ecosystems almost identically ($\rho = 0.98$ training, 0.98 inference): within this workload, faster *is* greener. The contrary finding is reproducible as an instrument artefact — the duration field of a widely used software estimator adds a constant 3.28s to any phase shorter than about 11s, inflating the apparent runtime of the shortest by up to 17 while leaving their energy correct.

5.5. RQ4: How much of this is the apparatus?

Every block in the campaign carries two independent energy readings over the same window. Figure 6 and Table 8 report what they say about each other.

On energy, the instruments agree. Over the whole campaign the ratio of CodeCarbon to counters is 1.0044 for the accelerator and 1.0070 for the CPU package, or 1.0049 for their sum: a disagreement of 0.5%. This is a useful and, we think, under-reported result. The literature’s concern with software estimators is

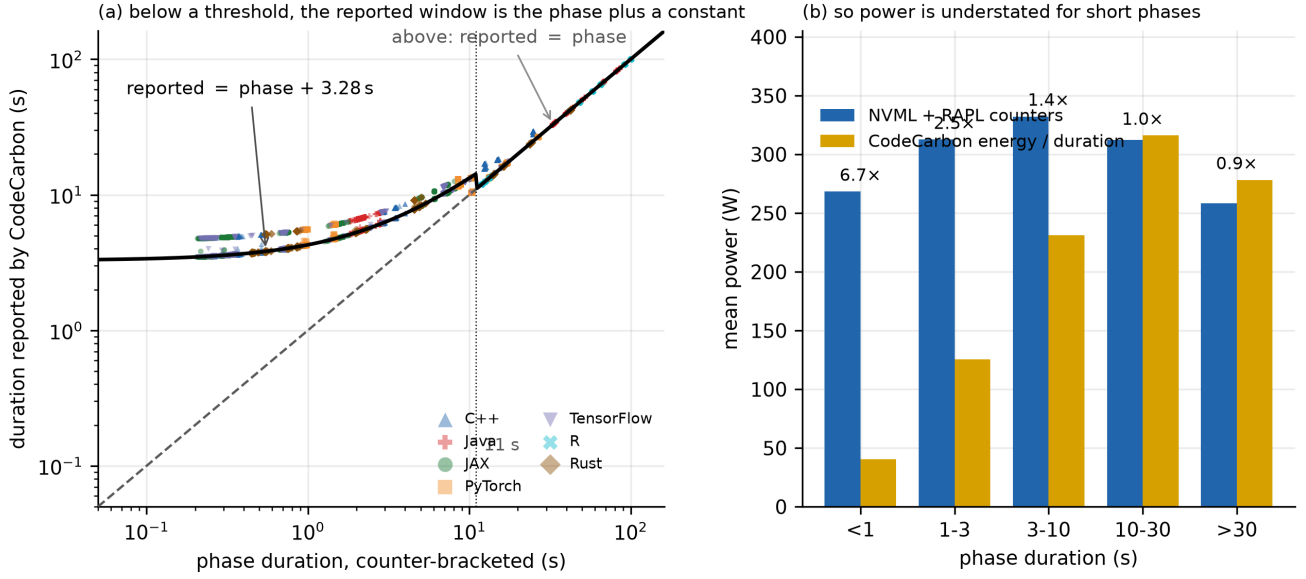


Figure 5: (a) The duration reported alongside each energy figure is not the interval the energy was accumulated over: it is the phase duration or a floor of about four seconds, whichever is larger. (b) Power obtained by dividing reported energy by reported duration is consequently understated for short phases and exact for long ones.

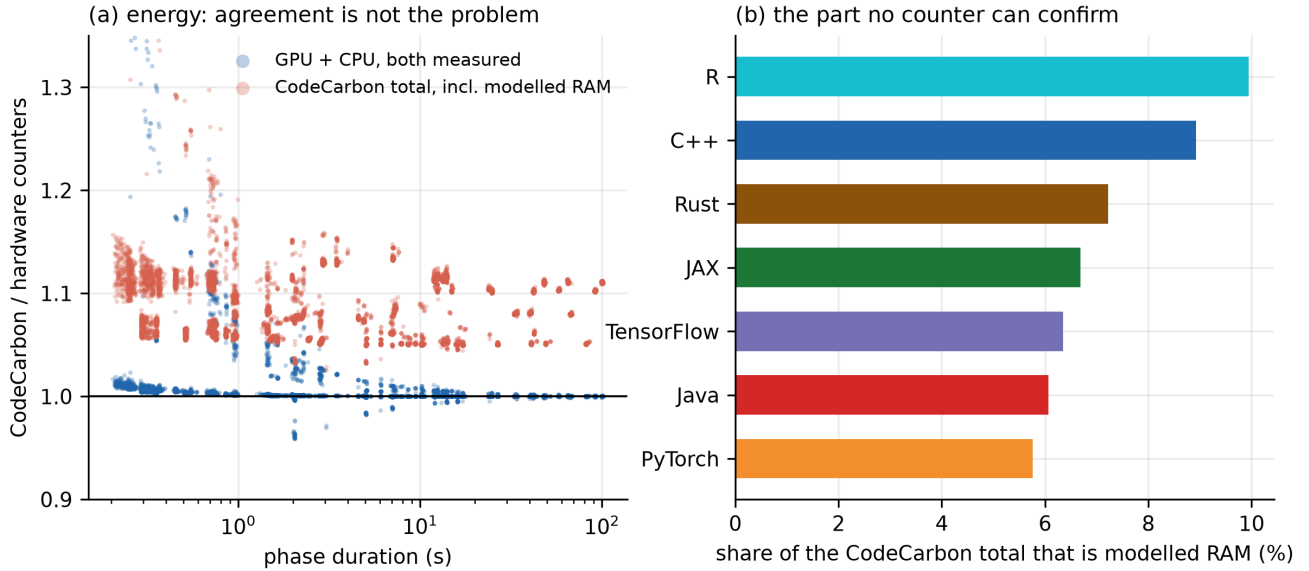


Figure 6: (a) Ratio of CodeCarbon to hardware counters, per block, against phase duration. The measured part agrees; the total does not, by exactly the modelled RAM term. (b) The share of each ecosystem's reported total that is modelled RAM.

Table 8

Agreement between the two instruments over 12600 blocks.

Quantity	Mean	5th pct	95th pct
GPU (NVML counter vs CodeCarbon pynvml sampling)	1.004	1.000	1.006
CPU package (RAPL counter vs CodeCarbon)	1.007	1.000	1.029
GPU + CPU, the measured part	1.005	1.000	1.011
CodeCarbon total incl. modelled RAM	1.084	1.050	1.130
RAM share of the CodeCarbon total	7.279	4.790	10.514

accuracy, and on this platform, over this workload, with both counters available, accuracy is not the problem.

That single figure hides where it comes from, and the structure matters more than the average. Agreement is a function of block length: above 30s the two instruments agree to 0.01 %, and below one second the mean error rises to 1.3 % with a worst case of 49 %. The transition sits at the sampling interval we configured, 1 s, which is consistent with the estimator accumulating counter deltas when it has samples to accumulate and interpolating from sampled power when it does not — though we can only observe the behaviour from outside, not the mechanism. The visible structure in Figure 6a is that transition.

The practical rule this implies is simple and, as far as we know, not stated anywhere: a measurement block must be several times the instrument's sampling interval, or the instrument is estimating rather than measuring it. That is an awkward constraint for exactly the case a cross-ecosystem study needs, because the blocks it most wants to compare — inference passes, and the epochs of the fastest stacks — are the short ones.

On what is not measured, they cannot agree. CodeCarbon's total exceeds the counters by 1.084, and the excess is entirely its RAM term — 7.3% of its reported total on average — which is derived from installed memory capacity rather than from memory activity. No counter on this platform can confirm or refute it. The number is not wrong so much as it is not a measurement, and it is added silently to a figure otherwise composed of measurements. On a machine with more installed RAM the same model contributes a larger share, which means the reported energy of an identical workload depends on how much memory the host happens to have.

The time between the phases belongs to the instrument. Energy is attributed only to the phases each stack brackets, and the stacks differ enormously in how much of a run that is. Measured over wall time, tracked time ranges from 44% (JAX) to 100% (R): more than half of a JAX run falls outside any measured block.

The natural reading is that the low-coverage stacks do a great deal of work between phases and are flattered by a per-phase comparison. We pursued that reading and it is wrong. The gap preceding each block is almost exactly the amount by which CodeCarbon's reported window exceeds the counter-bracketed phase: the two correlate at $r = 0.98$ over every block in the campaign, their medians differ by 0.23s, and the window excess accounts for 96% of all untracked time. Coverage does not track anything about the ecosystems except the length of their blocks (Table 9) — short blocks pay a fixed instrument cost more often, so JAX with a 3.2s median block is penalised where R with a 37s one is not.

The untracked time is the instrument holding its window open. It would not exist in an uninstrumented run, and charging it to the ecosystems would charge them for the cost of being measured. Per-phase energy is therefore the right quantity, and the spreads reported in Section 5.2 stand as they are.

What the observation does bear on is anyone planning a campaign of this shape. Under whole-machine tracking that overhead is real energy drawn from the wall and attributed to nobody, and its share grows as blocks get shorter — which is to say it grows exactly where the instrument is already least reliable.

Does the instrument change the answer? We recomputed the ecosystem ranking of each block under

Table 9

Tracked share of each run's wall time, and how much of the remainder is the instrument's own window. Coverage follows block length, not the ecosystem.

Ecosystem	Median block	Tracked share	Of the untracked rest, the instrument
JAX	3.2 s	44%	97%
TensorFlow	4.2 s	57%	100%
C++	5.2 s	62%	100%
PyTorch	7.8 s	78%	99%
Rust	8.0 s	80%	100%
Java	23.8 s	93%	99%
R	37.3 s	100%	15%

both instruments. They agree in 4 of 6 blocks. The disagreements are confined to adjacent pairs that the statistical tests do not separate, so the instrument does not reorder the substantive conclusions of RQ1 — provided energy, and not power or time derived from energy, is the quantity being compared.

RQ4 Findings: The Apparatus

Over 12600 doubly instrumented blocks, a widely used software estimator agrees with hardware energy counters to 0.5% on the quantity both measure — but that average is carried by the long blocks: it is 0.01% above 30s and 1.3% below one, so the estimator measures where the block outlasts its sampling interval and estimates where it does not. The ecosystem ranking is nonetheless unchanged in 4 of 6 blocks. The remaining failure modes lie elsewhere: a modelled RAM term contributing 7.3% of the reported total that no counter can check and that scales with installed memory rather than with use, and a reported duration that is a floor rather than a measurement. Both produce plausible numbers; neither is visible without a second instrument.

6. Discussion

Our study highlights that the energy efficiency of DL workloads is not only a matter of algorithmic or hardware design, but is also materially influenced by the *choice of language-framework ecosystem* (i.e. the adopted stack, including frameworks/libraries, run-times, and toolchains) [26, 27, 18, 21]. In this section, we interpret the results in light of their broader implications, and discuss lessons learned for different stakeholders.

6.1. Interpretation of Results

The ecosystem effect is large, and it is not about the kernels. Under a common measurement boundary, the same architecture trained on the same data for the same number of epochs costs between 7.8 and 19.1 more energy on one ecosystem than another. Three of the stacks in that comparison execute a byte-identical TorchScript module, so whatever separates them is not kernel quality: it is the binding, the data pipeline, and how much host-side work each stack does

per unit of accelerator work. That is an encouraging finding, because host-side overhead is the kind of thing an ecosystem maintainer can fix.

Ecosystem-level overheads amplify with workload scale. The absolute gap widens with dataset size and with model depth, so the practical relevance of the choice grows exactly where energy matters most. The mechanism cannot be isolated in a black-box study — runtime layers, input pipelines, execution engines and toolchains all differ — and we therefore report these as ecosystem-level effects rather than attributing them to a programming language.

Rankings are largely phase-consistent, contrary to what an estimator-based analysis suggests. Recomputed on counter-bracketed energy, training and inference rank the ecosystems alike in 6 of 6 blocks, with Spearman correlations between 0.93 and 1.00. This matters because the opposite claim — that the phases rank differently and so demand phase-aware stack selection — is a natural conclusion to draw from estimator output, and we can now say why: inference blocks in this workload are short enough to fall under the reported-window floor of Section 5.5, so an inference ranking derived from an estimator’s own duration is partly a ranking of floors. We do not claim phase-dependence never occurs; we claim that establishing it requires an instrument whose window is known.

Faster is greener, here. On measured durations, energy and time rank configurations at $\rho = 0.98$ (training) and 0.98 (inference). This contradicts a well-established finding in cross-language energy work [28, 30, 18], and we want to be careful about the scope of the contradiction. Those studies largely concern CPU-bound general-purpose benchmarks, where a language can trade instruction count against memory traffic and shift the energy-per-second substantially. In a GPU-bound DL workload the accelerator dominates and runs near a narrow band of power, so time and energy are nearly proportional almost by construction. The interesting result is not that we disagree, but that a study of this exact shape can produce either answer depending on which of two fields of the same CSV it treats as the duration.

Quality must be measured, not assumed. Under a fixed epoch budget the stacks converge to within 1.3 percentage points on Fashion-MNIST but separate by 13.1 points on CIFAR-100. Any efficiency claim on the harder datasets that does not report accuracy alongside energy is comparing stacks that did different amounts of learning. This is not a subtle correction: an ecosystem that converges more slowly looks efficient precisely because it accomplished less.

6.2. Implications for Stakeholders

Our findings carry different implications for different communities:

- **Researchers:** energy should be reported alongside accuracy, and both should be reported alongside the measurement boundary. Our stronger recommendation is procedural: a cross-stack study needs a written specification and a mechanism that enforces it. We found that every divergence we eventually corrected had produced a plausible number first, and that inspection did not catch them — executable checks did.
- **Practitioners:** ecosystem choice has a first-order effect on energy, and in this workload the same choice serves both phases, so the “one stack for training, another for serving” complexity is not obviously warranted. Efficiency must nonetheless be weighed against maintainability, hiring, library availability and time to delivery; an energy gap of this size is not a mandate when the cheaper stack has a tenth of the ecosystem around it.
- **Framework designers:** efficiency stems from the ecosystem as a whole, and the largest recoverable gaps we observe are in host-side work — data loading, dispatch and binding overhead — rather than in kernels. We would also ask for one specific thing from measurement tooling: report the interval over which energy was accumulated, and fail rather than substitute a model when a counter is unavailable. An instrument that refuses to run is more useful than one that quietly estimates.
- **Policy makers and funding agencies:** AI efficiency is multi-dimensional, and we caution against adopting any single metric as a target (Goodhart’s Law). Reporting requirements should ask for the measurement boundary and the replication count, not only the number: a figure without either is not auditable, and our own experience is that unauditable figures are wrong more often than not.

6.3. AI Requires Explicit Energy Measurement

Ecosystem selection is not a neutral engineering choice: it shifts DL energy consumption by a factor that is large, reproducible and, in the case of host-side overhead, addressable. That is the substantive result.

The methodological result is that establishing it is harder than it looks. Every defect we found in building this study produced numbers of the right order of magnitude, in the right units, with plausible relative ordering. A learning rate that differs between two framework APIs, an evaluation loop written one image at a time, a linker dropping a CUDA dependency so that a binary runs on the CPU, an estimator reporting a floor where a duration belongs — none of these announce themselves in the data. They are detectable only

by a second instrument, an executable specification, or a quality metric that the study happened to record.

We therefore make the artefacts as important as the findings. The specification, the shared measurement contract, the conformance checker and the raw dual-instrument records are all in the replication package, and the manuscript’s numbers are generated from them rather than transcribed. A reader who doubts a figure in this paper can regenerate it; a reader who doubts the apparatus can run the checks.

6.4. An Industrial Simulation of Deep Green AI Large-Scale Impact

To contextualize our empirical results in a hyperscale setting, we present an illustrative industrial scenario that translates our *relative* ecosystem-level energy differences into approximate energy and cost impacts. This scenario is not intended as a direct estimate for any specific production system, model family, or hardware stack; rather, it shows how small per-workload gaps can compound at scale.

Relative multipliers. For a given phase (training or inference) and ecosystem k , we define a phase-specific energy multiplier relative to any reference ecosystem *base* as:

$$m_k = \frac{\bar{E}_k}{\bar{E}_{\text{base}}},$$

where $\bar{E}_k$ is the mean energy measured under our experimental protocol for that phase. Accordingly, m_k should be interpreted as a *relative energy-per-work* factor under our setup, not as a universal constant.

Under the simplifying assumption that the ecosystem-level multiplier transfers to the same workload definition, for any baseline energy budget $\bar{E}_{\text{base}}$, the scaled energy is calculated as $\bar{E}_k = m_k \cdot \bar{E}_{\text{base}}$.

Training phase (illustrative scaling). From Table 3, moving from the most expensive ecosystem in a block to the cheapest changes training energy by 7.8–19.1. Taking a concrete case: if a *training* budget is $E = 1$ GWh on the most expensive stack of a block, the same fixed-budget run costs between 1/19.1 and 1/7.8 of that on the cheapest. At an electricity price of 100 e/MWh [7], the spread between the two ends of a single block is on the order of hundreds of thousands of euro per GWh of training. We stress that this is arithmetic on a relative multiplier, not a forecast: it assumes the multiplier transfers to a workload of a different size on different hardware, which is exactly the assumption our own measurements give us the least confidence in.

Inference phase (hyperscale workload, fixed work). Inference energy scales approximately linearly with served volume when stacks are compared at a fixed service target. Let E_{PyTorch} be the daily energy attributable to *inference* with PyTorch as the baseline.

Assuming $N = 100,000$ accelerators running continuously for 24 hours at an average inference-attributable draw of $P = 0.4$ kW each gives $E_{\text{PyTorch}} = 960$ MWh/day, or 96 000 € per day at 100 e/MWh. Applying the measured inference multipliers yields Table 10: JAX at 0.34 the baseline and R at 6.19.

Takeaway. Ecosystem-level energy differences, treated purely as relative multipliers under a uniform protocol, translate into large operational deltas at hyperscale. Two caveats bound that statement, and both come from this study rather than from convention. The multipliers were measured on a single-accelerator desktop at 32 32 input resolution, a regime in which data movement and dispatch weigh more heavily than they would at production scale, so they are more likely to overstate binding overhead than to understate it. And the whole calculation is a scaling of *relative* differences: had we reported the modelled component of a software estimator’s output as if it were measured, the same arithmetic would have produced a confident number from a quantity that depends on how much memory the host happens to have.

7. A catalogue of defect classes

Building this study meant first auditing an earlier campaign of the same design and then repairing every stack against the specification. Table 11 collects what that turned up. We report the defects as *classes* rather than as a list of bugs in one project, because each is a plausible mistake in any study of this shape, several are language or tool *defaults* rather than oversights, and most are invisible in the resulting measurements.

The last column is the one that matters. It asks whether a reader given only the released data — the tables, the CSVs, the plots — could detect the problem.

One row of that table deserves to be made concrete, because it is the class most likely to affect any study of this shape and the least likely to be noticed. Table 12 reports the training protocol *as implemented*, read from the source of an earlier campaign whose manuscript stated a single common learning rate.

Two stacks train at ten times the stated learning rate. Loader parallelism ranges from none to every hardware thread on the machine, and across the campaign it correlates with epoch duration at $\rho = *0.73$ ($p = 0.041$) over a 11.6 spread in duration and 9.9 in energy — which is to say that a column no one would think to report tracks the published ranking better than the programming language does. One stack normalises its inputs when no other does, so it is not even computing the same function. None of this is visible in an energy figure, and all of it is verifiable in about a page of code.

Three observations follow.

Only one of these is visible from the released data — and it is the one that changes no conclusion, since a constant factor leaves every ranking intact.

Table 10

Projected daily energy and cost for large-scale inference (10^5 accelerators, 24 h, electricity at 100 e/MWh), scaling the *relative* inference energy measured in this campaign with Python/PyTorch as the baseline. These are scenario arithmetic on measured multipliers, not forecasts.

Ecosystem	Relative energy	Energy (MWh/day)	Cost (€/day)	Saving vs PyTorch (€/day)
JAX	0.34	323	32 299	+63 701
TensorFlow	0.38	365	36 475	+59 525
C++	0.44	424	42 372	+53 628
Rust	0.85	821	82 059	+13 941
PyTorch (baseline)	1.00	960	96 000	—
Java	2.67	2 561	256 103	-160 103
R	6.19	5 941	594 104	-498 104

Everything that does change a conclusion requires reading the code, or a second instrument, or a quality metric the study has to have thought to record.

Several are defaults rather than oversights. The linker’s `--as-needed` behaviour, an estimator’s fallback power constants, a framework resolving CUDA wheels that do not match its build, a resize function that returns `uint8` because the images it was written for were `uint8`. None is a lapse in care. A study can be executed carefully and land on all of them.

Some of them are ours. Five entries were introduced while repairing the study, and each was caught by a mechanism rather than by attention: the conformance checks, the accuracy metric, or the second instrument.

One of the five deserves singling out, because it is not a mistake in the usual sense. Moving the input transform into the loader is required by S3. Switching evaluation to batched inference is required by S3 as well. Each change is correct, and each was reviewed. Together they left a private copy of the transform on the training path and removed it from the evaluation path, and the result trained a network on black images for thirty epochs while reporting a plausible falling loss. No amount of care applied to either change in isolation would have caught it; what caught it was a metric neither change was about. We report them because the alternative — a paper claiming its authors made no mistakes — would undercut the argument it is making. The argument is that enforcement beats discipline, and the evidence for it is that our own discipline was not sufficient either.

8. Threats to Validity

As with any empirical study, our work is subject to threats to validity. We group them conventionally, and note where a threat that would ordinarily be speculative is one we can quantify from the data.

Internal Validity. Whole-machine energy tracking means any concurrent work contaminates a measurement. The host was otherwise idle for the duration of the campaign; the blocks measured during development while a package download was running were discarded rather than corrected, and the protocol

amended to forbid it. We report this because it is an easy mistake to make and an invisible one to inherit.

Toolchain heterogeneity remains. CUDA versions across the 7 stacks range from 11.6 (Deeplearning4j, whose final release predates CUDA 12) to 12.8 (the pinned LibTorch shared by four ecosystems). This is not fully controlled and cannot be: an ecosystem is partly defined by the toolchain it supports. What we have changed is that it is now *bounded*. The four LibTorch stacks are pinned to one build, so the control group of Table 5 isolates binding overhead from backend differences, and a reader can see how much of the total spread survives that isolation.

Implementation drift between stacks was the dominant internal threat in studies of this shape, and it is the one this design attacks directly. Under specification S1 the LibTorch-based ecosystems train the same exported module rather than four reimplementations; S2 and S3 fix the optimisation and the data pipeline; and 57 automated checks verify all of it against the source. Two defects introduced during development were caught by those checks rather than by review. We do not claim the specification is complete — only that a divergence outside it now has to survive an executable check rather than an author’s attention.

Accelerator placement was verified rather than assumed. Every harness asserts the device at startup and refuses to fall back to the CPU. This is not hypothetical: linking a Rust binary against a CUDA LibTorch produces, by default, a binary that runs entirely on the CPU, because `--as-needed` drops the `torch_cuda` dependency that no Rust symbol references. Nothing in the energy data distinguishes that case from a correct one except its magnitude.

Construct Validity. Our construct is energy at a stated boundary, not “sustainability”. We report accelerator plus CPU-package energy read from hardware counters: a chip-level boundary that excludes power supply, fans and drives. We had no wall meter and therefore do not report a whole-system figure, and we would emphasise that a chip-boundary number must not be compared with a wall-meter number from another study.

The more interesting construct threat is one we can measure. A software estimator’s output is a mixture of

Table 11

Defect classes, with the measured effect of each. “Visible” asks whether a reader given only the released measurements could detect the problem. Entries marked < were introduced by us while repairing the study and caught by the checks or the accuracy metric, not by review.

Class	Measured effect	Visible
<i>Instrument</i>		
Unit mislabelling	Energy reported in kWh under a J label; a factor of $3.6 \cdot 10^6$	yes
Reported duration is not the accumulation window	Phase + 3.28s below 11s, phase above it; power understated 6.7 below one second, exact above ten	no
Modelled term inside a measured total	RAM energy from installed capacity, 7.3% of the reported total; scales with the host, not the workload	no
Mixed instrument versions	Versions 2.8.4 and 3.0.4 in one campaign: 231 W against 102 W of modelled host power for the same machine	no
Mixed tracking mode	1 of 8 stacks tracked per-process rather than machine-wide; its RAM term is 0.026~ of reported energy against 25~ to 55~ for the rest	no
Interpreter resolved from PATH	The instrument's version becomes a property of the shell; 5 of 8 stacks report a single constant CPU power, the signature of a modelled fallback	no
Unsynchronised tracker	Inference recorded as 0 J; training understated 24~	no
Host not idle	Whole-machine tracking charges unrelated downloads to the workload	no
<i>Implementation</i>		
Divergent hyperparameters	Two learning rates across eight stacks	no
Divergent loader parallelism	0 to 96 threads; $\rho = *0.73$ against epoch duration ($p = 0.041$) over a 11.6 spread	no
Divergent input shape	One stack at 0.26 another's conv1 input volume; 0.07 the energy	no
Divergent evaluation batching	Batch 1 against 128; 1.5–1.95 inference energy	no
Evaluation in training mode	Test accuracy 21~ against 86~; a different computation measured	no [£]
Resize discards a dtype conversion<	Image scaled to [0, 1], then resized by a function returning uint8; the scaling is silently thrown away	no [¤]
Two correct changes that are wrong together<	Moving the transform into the loader, and batching evaluation, each right alone; together they left a private copy of the transform on the training path only, which trained the network on black images for thirty epochs	no [£]
Channel order assumed, not checked<	[C,H,W] treated as [H,W,C]: the width cropped to three pixels and the result transposed	no [¤]
Ambiguous dataset layout	A class name containing / yields three different datasets	no
<i>Placement</i>		
Silent CPU fallback	Whole workload on the CPU after a CUDA wheel mismatch; one warning line	no
Linker drops the CUDA dependency	Same source, same LibTorch: Cpu against Cuda(0)	no
<i>Analysis</i>		
Collapsed runs counted as measurements	12 runs at chance accuracy averaged in with runs that trained; 20.2 against 1.3 points of apparent cross-stack spread	no [£]
Partial runs averaged as measurements<	A crashed run's fragment, 16 J against a neighbour's 300 kJ, produced ecosystem spreads of $2 \cdot 10^4$	no
Epochs treated as replications	Effective sample size 1 per configuration; intervals far too narrow	no

[£] invisible because no stack wrote its accuracy to disk; visible the moment one does.

[¤] invisible in energy data; here it happened to abort the run, but the channel-order defect would have trained quietly on a three-pixel sliver had the dtype defect not stopped it first.

Table 12

Training protocol as implemented in an earlier campaign of this design, read from source rather than from the manuscript. The manuscript stated one learning rate and one input scaling; the code contains two of each. Divergences in bold. Specification S2 and S3 now fix every column, and the conformance checks verify them against the source.

Ecosystem	Optimiser	LR	Loader mechanism	Threads	Input scaling
Python/PyTorch	Adam	1e-4	DataLoader workers	2	[0, 1]
Python/TensorFlow	Adam	1e-3	Keras generator	1	[0, 1]
Python/JAX	Adam	1e-4	Keras generator	1	[0, 1]
C++/LibTorch	Adam	1e-4	DataLoader workers	2	[0, 1]
Java/DL4J	Adam	1e-4	AsyncDataSetIterator	2	[0, 1]
R/torch	Adam	1e-4	dataloader workers	0	[0, 1]
Rust/tch	Adam	1e-3	rayon, all cores	96	mean/std
MATLAB/DLT	adam	1e-4	augmentedImageDatastore	1	[0, 1]

measured and modelled terms, and the mixture is not stated in the output. Over 12600 blocks, CodeCarbon agrees with the counters to 0.5 % on the terms both measure, while its total exceeds them by 1.084 — the difference being a RAM term derived from installed capacity, 7.3 % of the reported total here and necessarily larger on a large-memory host. A figure of that construction corresponds to no physical boundary, and its value depends on the host’s memory configuration rather than on the workload.

The estimator’s reported duration is a second construct threat, and the sharper of the two: below about 11 s it is the phase plus a constant 3.28 s rather than the accumulation window, so any power or energy–time quantity derived from it is biased as a function of phase length. We use counter-bracketed durations throughout for this reason.

External Validity. We studied two convolutional architectures and three image classification datasets at 32 × 32 resolution; results may not generalise to other model families, to transformers, or to higher-resolution workloads, where the balance between accelerator work and host-side work is different — and, given that host-side work is where we locate most of the observed spread, likely more favourable to the slower ecosystems than what we report.

The workload also does not saturate the accelerator. At this resolution and these dataset sizes the GPU runs well below its board power limit for much of each epoch, so the campaign measures data loading and dispatch more than it measures deep learning. This bounds the generality of the absolute figures in a specific direction: a saturating workload would compress the ecosystem spread.

The platform is a single-accelerator desktop rather than a datacentre node, which makes the host share of total energy smaller than it would be on a multi-socket server. And MATLAB, present in an earlier campaign of this design, is out of scope here because it cannot be brought into conformance with the specification; readers should not read its absence as a result about MATLAB.

Conclusion Validity. The submitted analyses of studies of this shape — including an earlier one of our own — treat the epochs of a single run as repeated measurements. They are not independent: epochs within a run share an initialisation, a JIT outcome, an allocator state and a thermal trajectory, so the effective sample size per configuration is one, confidence intervals are far too narrow and significance is inflated. Every interval and test reported here is computed on 5 independent runs per configuration, and all 210 runs were interleaved rather than executed consecutively so that machine-state drift is spread across conditions instead of aliasing onto one ecosystem.

Five repetitions is a small sample for a rank test, and we have preferred effect sizes (Cliff’s delta) and Holm-corrected pairwise comparisons over unadjusted p-values accordingly. Between-run variability is low — a median coefficient of variation of 0.44 % for training energy — which makes the design well powered for the effect sizes we report, and we note that low variance is a statement about the measurement rather than a guarantee that the measurement is of the right thing.

Where the sample is genuinely too small we say so rather than reach. The VGG-16 collapse rates of Section 5.2.1 range from 0 % to 50 % across ecosystems, and a χ^2 test on that table returns $p = 0.017$ — which would license a claim that ecosystems differ in robustness. The expected cell counts are far too small for that approximation; a permutation test on the same data gives $p = 0.12$. We report the permutation result and make no such claim. The difference between the two is exactly the kind of small-sample over-confidence a reader is entitled to be warned about.

Finally, the twenty Rust runs on Fashion-MNIST and Tiny ImageNet were executed in a separate, later time window than the rest of the campaign, after a defect in that stack’s image loader was corrected. They are therefore not interleaved with the other ecosystems, and machine-state drift between the two windows is not controlled for them. Given the between-run coefficients of variation we observe, we expect this to be small relative to the effects reported, but it is a

real asymmetry in an otherwise interleaved design and we flag it rather than absorb it.

9. Conclusion

This paper presented an empirical study of how *language-framework ecosystems* affect the energy efficiency of *Deep Learning* workloads, across 7 ecosystems, two convolutional architectures and three datasets, replicated 5 times per configuration and instrumented twice per measurement block.

Three findings stand out.

First, ecosystem selection substantially shifts operational efficiency: at a common chip-level boundary, training the same model on the same data costs between 7.8 and 19.1 more energy on one ecosystem than another, with between-run variability of order 0.44%. Because three of the ecosystems compared execute a byte-identical TorchScript module, this gap is not kernel quality: it is binding overhead, data movement and host-side dispatch — which is to say it is addressable by the people who maintain these stacks.

Second, the effect is energetic and not qualitative. All 7 stacks converge to within 1.3 percentage points on Fashion-MNIST under the same epoch budget. They are doing the same work at very different cost.

Third — and this is the finding we did not expect — the apparatus determines more of the answer than the ecosystems do, in one specific and correctable way. Over 12600 doubly instrumented blocks, a widely used software estimator agrees with hardware energy counters to 0.5% on energy. But the duration it reports beside that energy is the phase plus a constant 3.28s, not the interval the energy was accumulated over, so power derived from the two fields is understated by up to 6.7 for short phases and exact for long ones. Recomputed on measured durations, energy and time rank ecosystems at $\rho = 0.98$: in this workload, faster *is* greener, and the widely reported contrary result is reproducible here as an instrument artefact.

We draw a procedural conclusion from that. A cross-ecosystem energy study needs a written specification, a mechanism that enforces it, and a second instrument — for the same reason that a performance benchmark needs a stated warm-up policy. Every defect we corrected in building this study produced a plausible number first. The specification, the shared measurement contract, the 56-check conformance checker and the raw dual-instrument records are part of the replication package, and every figure in this paper is generated from them.

10. Future Work

Several directions remain open.

First, the workload should be extended beyond CNNs, to transformers and multimodal architectures, and to input resolutions that saturate the accelerator.

Our workload does not: the GPU runs well below its board power limit for much of each epoch, which is precisely the regime in which host-side overhead — the component we find dominates the ecosystem gap — weighs most. Establishing how the gap narrows as the accelerator saturates is the single most useful follow-up, and it is a measurement, not a speculation.

Second, testing on heterogeneous hardware (server-class accelerators, TPUs, embedded and edge devices) would establish how much of the ranking is a property of the stacks and how much of this host. The hardware–software stack should be treated as part of the ecosystem definition rather than as a nuisance parameter.

Third, a white-box layer complementing this black-box design would localise where the host-side overhead originates: loader thread scheduling, host-to-device transfer, per-kernel launch overhead, allocator behaviour. We can say that three stacks running identical kernels differ by a large factor; we cannot yet say which line of the binding is responsible.

Fourth, the instrument characterisation invites replication. We report the reported-window floor of one estimator on one platform. Whether other estimators share it, whether it varies with sampling interval and tracking mode, and how many published energy–time results rest on it, are answerable questions, and the method — bracket every block with hardware counters and compare — is cheap.

Finally, a wall-meter reference would let the chip-level boundary used here be related to whole-system energy, which is what an operator pays for. We report a chip boundary consistently and decline to extrapolate; closing that gap requires hardware we did not have.

11. Carbon Footprint of This Study

The campaign reported here consumed 48.7 MJ (13.5 kWh) at the accelerator and CPU package across 210 runs. At the grid carbon intensity CodeCarbon assigns to this location, that corresponds to a few kilograms of CO₂e — modest against industrial training, and reported here for the same reason we ask others to report it: an energy figure without a stated boundary and a stated scope is not auditable. The figure above excludes development, failed runs and the earlier campaign that motivated this design; including them would roughly double it.

12. Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the authors used generative AI assistants (ChatGPT and Claude) in three distinct capacities, which we separate because they carry different weight.

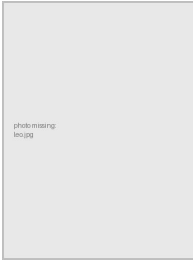
For *writing*, the tools were used to improve the readability of the manuscript text. For *software engineering*, they were used to repair and refactor the experimental implementations, to write the shared measurement bridge, the conformance checker and the analysis pipeline, and to diagnose defects in the ecosystems' data loaders and build configurations. For *analysis*, they were used to write the scripts that produce the tables and figures; the scripts, not the assistants, produce the numbers, and the numbers are injected into this manuscript from those scripts rather than transcribed.

No experimental measurement was produced, adjusted or selected by an AI assistant. The authors reviewed and edited all generated content and code, verified the findings against the raw data, and take full responsibility for the content of the article.

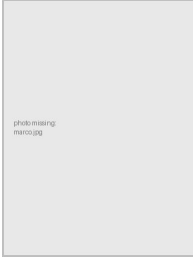
References

- [1] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 3645–3650.
- [2] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," *arXiv preprint arXiv:1910.09700*, 2019.
- [3] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big?" in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021, pp. 610–623.
- [4] J. Xu, W. Zhou, Z. Fu, H. Zhou, and L. Li, "A survey on green deep learning," *arXiv preprint arXiv:2111.05193*, 2021.
- [5] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "Carbon emissions and large neural network training," *arXiv preprint arXiv:2104.10350*, 2021.
- [6] A. S. Luccioni and A. Hernandez-Garcia, "Counting carbon: A survey of factors influencing the emissions of machine learning," *arXiv preprint arXiv:2302.08476*, 2023.
- [7] L. H. Kaack, P. L. Donti, E. Strubell, G. Kamiya, F. Creutzig, and D. Rolnick, "Aligning artificial intelligence with climate change mitigation," *Nature Climate Change*, vol. 12, pp. 518–527, 2022.
- [8] D. Patterson, J. Gonzalez, U. Hölzle, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. R. So, M. Texier, and J. Dean, "The carbon footprint of machine learning training will plateau, then shrink," *Computer*, vol. 55, no. 7, pp. 18–28, 2022.
- [9] A. S. Luccioni, S. Viguier, and A.-L. Ligozat, "Estimating the carbon footprint of BLOOM, a 176B parameter language model," *Journal of Machine Learning Research*, vol. 24, no. 253, pp. 1–15, 2023, arXiv:2211.02001.
- [10] C. Banbury, V. J. Reddi, P. Torelli, J. Holleman, N. Jeffries, C. Kiraly, P. Montino, D. Kanter, S. Ahmed, D. Pau, U. Thakker, A. Torrini, P. Warden, J. Cordaro, G. Di Guglielmo, J. Duarte, S. Gibellini, V. Parekh, H. Tran, N. Tran, N. Wenxu, and X. Xuesong, "MLPerf Tiny benchmark," in *Proceedings of the NeurIPS 2021 Track on Datasets and Benchmarks*, 2021, arXiv:2106.07597.
- [11] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," *Journal of Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.
- [12] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.
- [13] R. Verdecchia, J. Sallou, and L. Cruz, "A systematic review of Green AI," *WIREs Data Mining and Knowledge Discovery*, vol. 13, no. 4, p. e1507, 2023.
- [14] V. Mehlin, S. Schacht, and C. Lanquillon, "Towards energy-efficient deep learning: An overview of energy-efficient approaches along the deep learning lifecycle," *arXiv preprint arXiv:2303.01980*, 2023.
- [15] C. E. Tripp, J. Perr-Sauer, J. Gafur, A. Nag, A. Purkayastha, S. Zisman, and E. A. Bensen, "Measuring the energy consumption and efficiency of deep neural networks: An empirical analysis and design recommendations," *arXiv preprint arXiv:2403.08151*, 2024.
- [16] T. Yarally, L. Cruz, D. Feitosa, J. Sallou, and A. van Deursen, "Uncovering energy-efficient practices in deep learning training: Preliminary steps towards green AI," in *2023 IEEE/ACM 2nd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2023, pp. 25–36, arXiv:2303.13972.
- [17] L. Cruz, J. P. Fernandes, M. H. Kirkeby, S. Martínez-Fernández, J. Sallou, H. Anwar, J. Bogner, J. Castaño, F. Castor, D. Feitosa, A. González, P. Lago, H. Muccini, A. Oprea, P. Rani, J. Saraiva, F. Sarro, R. Selvan, K. Vaidhyathan, R. Verdecchia, and I. P. Yamshchikov, "Greening AI-enabled systems with software engineering: A research agenda for environmentally sustainable AI practices," *ACM SIGSOFT Software Engineering Notes*, 2025, arXiv:2506.01774.
- [18] R. Pereira, M. Couto, F. Ribeiro, R. Rua, J. Cunha, J. P. Fernandes, and J. Saraiva, "Ranking programming languages by energy efficiency," *Science of Computer Programming*, vol. 205, p. 102609, 2021.
- [19] S. Georgiou, M. Kechagia, and D. Spinellis, "Analyzing programming languages' energy consumption: An empirical study," in *Proceedings of the 21st Pan-Hellenic Conference on Informatics (PCI)*, 2017.
- [20] R. Pereira, M. Couto, F. Ribeiro, R. Rua, J. P. Fernandes, J. Saraiva, and J. Cunha, "Energy efficiency across programming languages: how do energy, time, and memory relate?" in *Proceedings of the 10th ACM SIGPLAN International Conference on Software Language Engineering (SLE)*, 2017, pp. 256–267.
- [21] A. Gordillo, C. Calero, M. . Moraga, F. García, J. P. Fernandes, R. Abreu, and J. Saraiva, "Programming languages ranking based on energy measurements," *Software Quality Journal*, vol. 32, no. 4, pp. 1539–1580, 2024.
- [22] S. Nanz and C. A. Furia, "A comparative study of programming languages in Rosetta Code," in *Proceedings of the 37th International Conference on Software Engineering (ICSE)*, vol. 1, 2015, pp. 778–788, arXiv:1409.0252.
- [23] T. B. Chandra, P. Verma, and A. K. Dwivedi, "Impact of programming languages on energy consumption for sorting algorithms," in *Software Engineering*, ser. Advances in Intelligent Systems and Computing, vol. 731. Springer, Singapore, 2019.
- [24] S. Mahadevappa and S. Figueira, "Energy-efficient programming languages for mobile applications," in *2021 IEEE Global Humanitarian Technology Conference (GHTC)*, 2021.
- [25] S. Maleki, C. Fu, A. Banotra, and Z. Zong, "Understanding the impact of object oriented programming and design patterns on energy efficiency," in *2017 Eighth International Green and Sustainable Computing Conference (IGSC)*, 2017, pp. 1–6.

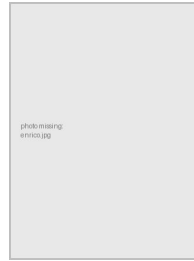
- [26] S. Georgiou, M. Kechagia, T. Sharma, F. Sarro, and Y. Zou, "Green AI: Do deep learning frameworks have different costs?" in *Proceedings of the 44th International Conference on Software Engineering (ICSE)*, 2022, pp. 1082–1094.
- [27] N. Marini, L. Pampaloni, F. Di Martino, R. Verdecchia, and E. Vicario, "Green AI: Which programming language consumes the most?" in *2025 IEEE/ACM International Workshop on Green and Sustainable Software (GREENS)*, 2025, arXiv:2501.14776.
- [28] M. Weber, C. Kaltenecker, F. Sattler, S. Apel, and N. Siegmund, "Twins or false friends? A study on energy consumption and performance of configurable software," in *Proceedings of the 45th International Conference on Software Engineering (ICSE)*, 2023, pp. 2098–2110.
- [29] J. Dodge, T. Prewitt, R. Tachet des Combes, E. Odmark, R. Schwartz, E. Strubell, A. S. Luccioni, N. A. Smith, N. DeCario, and W. Buchanan, "Measuring the carbon intensity of AI in cloud instances," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022, pp. 1877–1894.
- [30] R. Rua and J. Saraiva, "A large-scale empirical study on mobile performance: energy, run-time and memory," *Empirical Software Engineering*, 2024.
- [31] S. Aquino-Brítez, A. Ortiz, J. Ortega, J. J. Escobar, and D. Aquino-Brítez, "Towards an energy consumption index for deep learning models: A comparative analysis of architectures, GPUs, and measurement tools," *Sensors*, 2025, compares OpenZmeter hardware sensors against CarbonTracker and CodeCarbon on AlexNet, ResNet-18, VGG-16 and Swin Transformer.
- [32] L. Bouza, A. Bugeau, and L. Lannelongue, "How to estimate carbon footprint when training deep learning models? A guide and review," *Environmental Research Communications*, vol. 5, no. 11, p. 115014, 2023.
- [33] R. Fischer, "Ground-truthing AI energy consumption: validating CodeCarbon against external measurements," *it - Information Technology*, 2025, reports estimation errors of up to 40% for established software estimators against meter ground truth.
- [34] E. Paula, J. Soni, H. Upadhyay, and L. Lagos, "Comparative analysis of model compression techniques for achieving carbon efficient AI," *Scientific Reports*, vol. 15, p. 23461, 2025.
- [35] T. Almog, "An analysis of optimizer choice on energy efficiency and performance in neural network training," *arXiv preprint arXiv:2509.13516*, 2025, vERIFY. The citing sentence calls for an optimizer efficiency benchmark that uses CodeCarbon, and this paper is exactly that – 360 runs over eight optimizers, tracked with CodeCarbon. But the key says "patterson" and the sole author is Almog, so either the key was mistyped or a different paper was intended.
- [36] S. Luccioni, Y. Jernite, and E. Strubell, "Power hungry processing: Watts driving the cost of AI deployment?" in *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024, pp. 85–99, vERIFY. The citing sentence calls for a large-scale evaluation of large language models instrumented with CodeCarbon; this is the best-known 2024 paper of that description, but the key is generic and no title is recoverable from it.
- [37] K. C. Aashish *et al.*, "Towards eco friendly cybersecurity: machine learning based anomaly detection with carbon and energy metrics," *arXiv preprint*, 2025, instruments Logistic Regression, Random Forest, SVM, Isolation Forest and XGBoost with CodeCarbon and defines an F1-per-kilowatt-hour eco-efficiency index.
- [38] R. Feldt and A. Magazinius, "Validity threats in empirical software engineering research – an initial survey," in *Proceedings of the 22nd International Conference on Software Engineering & Knowledge Engineering (SEKE)*, 2010, pp. 374–379.
- [39] A. Ampatzoglou, S. Bibi, P. Avgeriou, M. Verbeek, and A. Chatzigeorgiou, "Identifying, categorizing and mitigating threats to validity in software engineering secondary studies," *Information and Software Technology*, vol. 106, pp. 201–230, 2019.
- [40] D. I. K. Sjøberg and G. R. Bergersen, "Construct validity in software engineering," *IEEE Transactions on Software Engineering*, vol. 49, no. 3, pp. 1374–1396, 2023.
- [41] R. Verdecchia, E. Engström, P. Lago, P. Runeson, and Q. Song, "Threats to validity in software engineering research: A critical reflection," *Information and Software Technology*, vol. 164, p. 107329, 2023.
- [42] V. R. Basili, G. Caldiera, and H. D. Rombach, "The goal question metric approach," in *Encyclopedia of Software Engineering*, vol. 1. John Wiley & Sons, 1994, pp. 528–532.
- [43] T. D. Cook and D. T. Campbell, *Quasi-Experimentation: Design and Analysis Issues for Field Settings*. Boston: Houghton Mifflin, 1979.
- [44] D. T. Campbell and J. C. Stanley, *Experimental and Quasi-Experimental Designs for Research*. Boston: Houghton Mifflin, 1963.
- [45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778, arXiv:1512.03385.
- [46] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *3rd International Conference on Learning Representations (ICLR)*, 2015, arXiv:1409.1556.
- [47] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017.
- [48] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.



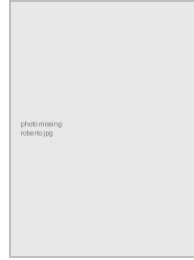
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